Hearing Recording Access

An alternative approach.

1. Requirements

1. Users who share hearing recordings can be managed in the staff reference data UI.
2. Users can be given permission to share recordings for many services, either individually or aggregated (e.g. 50 separate things can be granted, each giving access to share recordings for 1 service, or 10 things can be granted, each giving access to share recordings for ~5 services).
3. Users will only be able to share recordings - i.e. have access to a specific CCD event - for services where they have been given permission.

2. Constraints

1. "Hearing recordings" is a CCD jurisdiction and a single case type. Services do not have "Hearing recording" case types within their own jurisdiction.

3. Background

1. In the staff UI, users can be associated with "services" (aka "areas of work"). These are currently configured with a static list in EXUI, which associates each selectable service with one or more service codes. The service codes are currently standard MRD service codes - e.g. BFA1, blocked URL AAA7, etc.
2. The staff UI supports the definition of skills for users, which act in the same way as judicial authorisations, affecting the access the user is given to cases and tasks.
3. The CCD access control framework supports a case being given a single, service-defined "Case Access Category" which is matched against role configurations in the case definition to turn individual role access profile mapping rules on and off.

4. Solution

4.1 Define a Hearing Recordings "service" in staff reference data

The static list of selectable services maintained by EXUI is extended with an entry and a code for hearing recordings, e.g.:

```
{service: HRS, serviceCodes: ["HRS1"]}
```

(Assuming the CCD jurisdiction code is HRS.)

4.2 Define a "skill" for each recording sharing permission required

(See the note below on skill vs authorisation.)

For each separate assignable permission, a skill is defined in staff reference data, and associated with the HRS jurisdiction.

e.g. `"Share Civil hearing recordings" : "SKILL:HRS1:SHARE:CIVIL"`

There can be 50 skills created for fine-grained sharing permissions, or (preferable) a smaller number representing aggregated permissions.

> ⚠ It is not yet clear whether 'share' and 'listen' skills are needed separately (given that there may be separate roles) - it may be that only `SKILL:HRS:CIVIL` is needed, for example, in conjunction with the roles available.

4.3 Define an HRS case access category corresponding to each skill

e.g.

Skill Code	Case Access Category
`SKILL:HRS1:SHARE:CIVIL`	`HRS-CIVIL`
`SKILL:HRS1:SHARE:DIVORCE`	`HRS-DIVORCE`
etc.	

4.4 Create role assignments for HRS users

Create Organisational Role Mapping rules which give a suitable role assignment (e.g. `hrs-administrator`) to users who are configured with the HRS service, copying across their skills as usual in mappings.

Note that it may also be necessary / advisable to create a job title for these users - or one or more existing job titles may be suitable for use.

4.4 Assign the correct case access category to each HRS case

See Required Data Items for Business Outcomes#CaseAccessManagement for details.

This will be implemented in the HRS case type callbacks, setting the category based on the service the recording is for.

e.g.:

```
    {
  "data": {
    "caseAccessCategory": "HRS-CIVIL"
  }
}
```

4.5 Configure RoleToAccessProfiles with **both** skills and case access categories

CCD RoleToAccessProfiles case definition records specify both a case access category and a skill ("Authorisation"), e.g.:

```
      {
    "CaseTypeID": "HRS",
    "RoleName": "hrs-administrator",
    "Authorisation": "SKILL:HRS1:SHARE:CIVIL",
    "CaseAccessCategories": "HRS-CIVIL",
    "AccessProfiles": "sharing",
  }
```

According to the specification for RoleToAccessProfile entries, the above record will be interpreted:

- IF the user has a role assignment with the role name `hrs-administrator` ...
- ... AND the **user** has the skill `SKILL:HRS1:SHARE:CIVIL` ...
- ... AND the **case** has a case access category value starting `HRS-CIVIL` ...
- ... THEN the user is given the `sharing` access profile on the case.

The `sharing` access profile can include any necessary permissions to access and update case data and events.

4.6 How it works

Based on the staff reference data and the ORM mapping rule, a user will be given a role assignment with properties such as:

```
    roleName: "hrs-administrator"
authorisations: ["SKILL:HRS1:SHARE:CIVIL"]
attributes:
        jurisdiction: "HRS"
```

HRS cases containing recordings for Civil cases will have the appropriate access category:

```
    data:
    caseAccessCategory: "HRS-CIVIL"
```

The role to access profile mapping will only give users with the corresponding skill access to the case (more accurately: the specific profile which allows them to run the event to share the case - they may also get read access through other roles, for exmaple):

```
    {
  "CaseTypeID": "HRS",
  "RoleName": "hrs-administrator",
  "Authorisation": "SKILL:HRS1:SHARE:CIVIL",
  "CaseAccessCategories": "HRS-CIVIL",
  "AccessProfiles": "sharing",
}
```

The above is based on using case categories and authorisations/skills together. According to the specifications for how each of these is intended to work, the functionality should be OK. However, these have not been used together before, or tested together, so there may be issues encountered. A spike to confirm that they work together well is advisable before committing significant resources to this approach.

5. Points for discussion

5.1 UI complexity

This approach appears to deal well with the problems identified around UI complexity and changes:

- Only one additional "service" option is needed.
- Multiple fine-grained options (10 or 50) appear as skills, and so are only shown if the hearing recordings "service" has been selected - otherwise they do not clutter the UI.

5.2 Staff reference data "service" and MRD "service"

The selectable services in SRD currently correspond to (one or more) service codes, which reference a particular level of the MRD organisational structure.

"Hearing Recordings" is (presumably) not currently part of that structure, and a code would be needed to represent it, which might be perceived as an additional, locally-defined code within the MRD organisational structure namespace.

However, just because the current set of codes in SRD point to MRD organisational units does not mean that this was a design decision. The driver for the values available should be seen as "coherent subsets of the business that we may want to associate a user with" - this is a superset of the MRD organisational units (and might include organisational units at other levels of the MRD hierarchy). We should not necessarily see "hearing recordings" as a new MRD organisational unit, or a locally-defined one (with the complexities that introduces for allocating codes): it is better to see the values held in SRD as a wider set, which includes some MRD organisational unit codes, and possibly some other forms of identifier. That may mean that we should add scoping to the SRD data model, so that we know what namespace each assigned code is from.

5.3 "Skills are not permissions"

A skill such as `SKILL:HRS1:SHARE:CIVIL` is not a skill in the same sense as skills created by other services, which reflect more a qualification to do a set of tasks than permission / authorisation.

However, staff skills are exactly the same under-the-covers as judicial authorisations - it's just that the term "skill" fits the nature of the staff requirement better.

If there is any objection to referring to the HRS permissions as "skills", we could extend the staff reference data UI to have both a "skills" sections and an "authorisations" section: each section would maintain a separate subset of the same underlying collection. For each skill / authorisation, we would need to know which it was, which would mean either extending the SRD data model, or using a naming convention - for example, **`AUTH`**`:HRS1:SHARE:CIVIL` , instead of **`SKILL`**`:HRS1:SHARE:CIVIL`.

5.4 Number of skills per user

A fine-grained decomposition of HRS skills is possible, with a separate skill for each service for which recordings are processed. However, handling skills adds complexity to the CCD access control layer, and there is a strong recommendation to minimise the number of skills an individual user is given, aiming for a maximum of 10 skills. Decisions on the granularity of permissions to be used should take this into account, considering the users who have the widest range of services for which they share recordings.

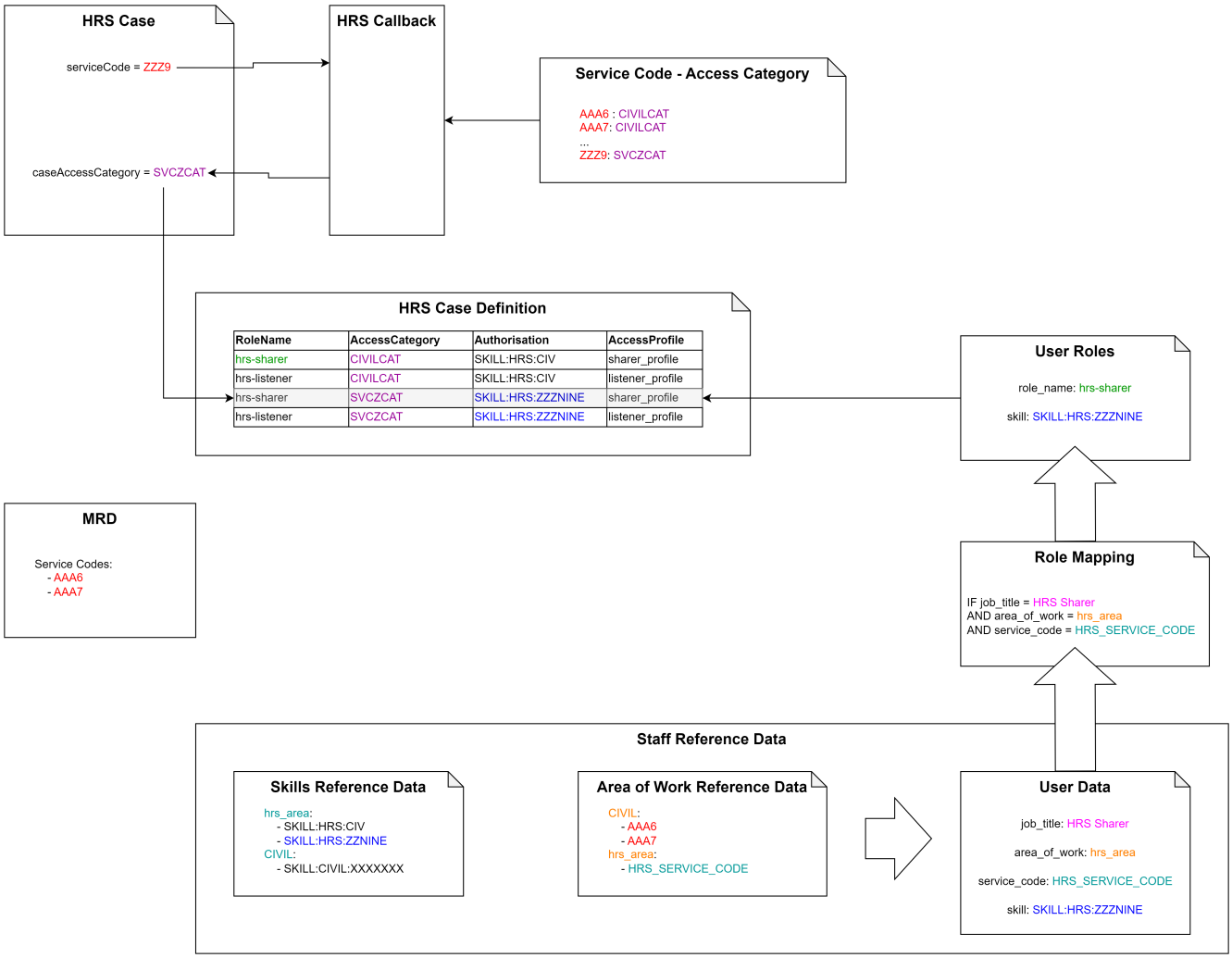
5.5 Skill / authorisation granularity

There is no need to impose uniformity of the granularity of skills / authorisations. There could be a single authorisation for the whole of one service (`SKILL:HRS1:SHARE:PRIVATE-LAW`), and multiple authorisations for another (`SKILL:HRS1:SHARE:CIVIL:SPEC, SKILL:HRS1:SHARE:CIVIL-UNSPEC`). The fundamental requirement is that it must be possible to tag the case with a case access category corresponding to the skill, which depends on the data captured from the users about each recording.

It is also possible for coarse-grained and fine-grained authorisations to co-exist, because case category matching is based on a starts-with match, not an exact match. By structuring the case access category appopriately, one user could have permissions for "all family cases", while another just has permissions for "private law" cases.

6. Reference data and code values

6.1 Categories of code values and their usage



The diagram is a schematic of how reference data and codes are used and matched throughout the process:

(Note that all codes are internal to CFT, except for the MRD service codes.)

MRD Service Codes (red)

Used in the HRS case data to associate the case with the appropriate service.

Used by the service callback to map the case to an appropriate Access Category Code.

Access Category Codes (purple)

Internal to the HRS case definition and code.

Set as the caseAccessCategory in the case data.

Referenced by RoleToAccessProfiles mappings in the case definition.

Used to filter the applicable RoleToAccessProfiles mappings to the ones matching the case access category of the case.

Skill Codes (blue)

Defined in staff reference data to represent a skill / authorisation a user can possess, and linked to areas of work.

Copied into the user's role assignments through the role mapping process.

Referenced by RoleToAccessProfiles mappings in the case definition.

Used to filter the applicable RoleToAccessProfiles mappings to those for which the user has an applicable skill.

Role Names (green)

Defined by the role assignment service.

Assigned to users by the organisational role mapping process.

Used to filter RoleToAccessProfiles mappings to those for which the user has the right role.

Job Titles (pink)

Defined in the staff reference data service, so that admins can select "what this user does / is responsible for".

Referenced in role mapping rules, to ensure that users get the appropriate roles for their responsibilities.

(Note: there is an ID for each job title, which is used in practice.)

Area of Work Codes (orange)

Defined in the staff reference data service, representing "parts of the business within which a user can be given responsibility".

Current values closely shadow CCD case types, but this should not be seen as a requirement - it is just a reflection of the requirements implemented to-date.

Included in the user information returned to clients such as the Organisational Role Mapping (ORM) service.

Can be used by the ORM in conditions of role assignment rules (but no rules currently reference areas of work, only service codes).

Service Codes in Staff Reference Data (red)

Defined in MRD.

Mapped to areas of work, to provide standard codes for the HMCTS organisational units a user works in.

Included in the user information returned to clients such as the Organisational Role Mapping (ORM) service.

Used by the ORM in conditions of role assignment rules.

6.2 Areas of work and service codes

It is likely that clients expect to find an HMCTS level 4 organisational unit code in the serviceCode attribute of staff data, and may validate the data or otherwise use it in ways that could break if we introduce codes not mastered in MRD as level 4 codes.

To associate a user with the HRS "service" and provide data to the ORM to drive the mapping rules therefore needs either:

1. Additional code data, similar to service codes, but stored and provided in separate attributes, to avoid confusion; or
2. The ability to have an area of work without any service codes: ORM rules are then implemented based solely on the area of work.

(2) seems more straightforward, since both options will lead to the need for an area of work to be created without any corresponding service codes.

Question: are there constraints in (a) the staff UI; (b) the staff reference data service logic; which require at least one service code to be defined for an area of work?